PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

JOINT INSPIRING EXAMINATIONS FOR PRIVATE SECONDARY SCHOOLS

(JIEPSS)

FORM THREE JOINT TERMINAL EXAMINATION

032 **CHEMISTRY**

Time: 03 Hours **May/june , 2025**

Instructions

1. This paper consists of section A, B and C with a total of **eleven (11)** questions.
2. Answer all question in sections A and B and any **Two (2**) questions from section C
3. Section A carries 16 marks, section B carries 54 marks and section C carries 30 marks.
4. Any unauthorized materials are **not** allowed in the examination room
5. Write **your examination number/name** on every page of your answer sheets
6. The following constant may be used

   Atomic masses H=1 O=16 Mg=24 S=32

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i)-(x) choose the most correct answer from the given alternatives and write its letter beside the item number in the answer sheet provided

(i) Many chemical processes involve the substances in either the liquid or gaseous or gaseous state. Which of the following statement is correct?

A. A fixed mass of has a fixed volume at room temperature
B. A specific mass of a liquid has a fixed volume at room temperature
C. $25cm^3$ of gas contains more particles than $25cm^3$ of liquid
D. Liquid particles placed inside a sealed container have fixed position

(ii) A student noted the following observation in his laboratory notebook. "The particles moved slowly from an area of high concentration to an area of low concentration". Which process is being described

A. Diffusion of a substance in liquids
B. The particle of solid during melting
C. A liquid being frozen
D. Diffusion of a substance through air

(iii) Substance can be broadly classified into three categories which are elements, compounds and mixture. Which of the following statements is corrects?

A. Elements contains different atoms
B. Compounds contains elements which are not chemically combined
C. Mixture consists elements that are chemically combined together
D. A mixture contains elements and compounds

(iv) Elements X and Y combines to form an ionic compound.Atoms of X have more protons of atoms Y. Atoms of Y have more valence electrons than atoms of X. Which statements is correct

A. Ions of X are negatively charged
B. Atoms of X have more electrons shell than that of atoms Y
C. X and Y are in the same period of the periodic table
D. X and Y are in the same group in the periodic table

(v) When a burette is used in the laboratory its calibration starts from top because

A. It measures the volume of liquids remained
B. It measures the volume of liquids used
C. It measures the volume of acid
D. It measure the volume of liquid added

(vi) The mass of sulphuric acid present in $350cm^3$ of 0.1M aqueous solution is

A. 3.43g
B. 2.94g
C. 9.8g
D. 4.9g

(vii) A general term used to express a mixture of two or more metals is called

A. Alloy
B. Isotopy
C. Isotope
D. Amorphous

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(viii) Technician prefer to use blue flame in heating and boiling because

A. Its bright and non-soot
B. Its light and non-soot
C. It's very hot and non-sooty
D. It's not expensive

(ix) Is the laboratory apparatus which is used for carrying burning substance during experiments

A. Deflagrating spoon
B. Spatula
C. Crucible
D. Woulf's bottle

(x) Which of the following cannot be used to prepare hydrogen gas

A. Electrolysis of water
B. Reaction of water and carbon at room temperature
C. Reaction of water with hot metals
D. Reaction of dilutes acids with zinc granules

2. Match the item in list A with the corresponding answers in list b by writing the letter beside the item number on the answer sheet provided

List A	List B
(i) Foam	A. Potassium acetate
(ii) Wet chemical	B. Bromochloro-difluoro-methane
(iii) Halon	C. Proteins and Flouro-proteins
(iv) Dry chemical	D. Ordinary tap water
(v) ABC	E. Non-flammable carbon dioxide
(vi) APW	F. Monoammonium phosphate
	G. Fine sodium carbonate powder pressurized with nitrogen

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Destructive distillation of coal is generally carried out in two types of kiln. These are beehive kiln and the Otto Hoffman kiln. Distinguish this two types of kiln under the following criteria
(i) Amount of fuel produced (one point)
(ii) Cost of the method used (one point)
(iii) More beneficial method (one point)

(b) Ernest Rutherford reason that if Thomson's model was correct then the mass of an atom was evenly distributed throughout. But the experiment done by Rutherford found that the mass of an atom was actually concentrated in the central core. As a form three student explain three findings of Rutherford touched subatomic particles.

4. (a) Hutson and Hinju went to the laboratory and found out that it was filled with smell of laboratory gas. They later discovered that the smell was because one gas tap at the corner of the laboratory had been left open
(i) Use kinetic theory of matter, explain why smell of the gas reached all parts of the laboratory
(ii) What is the name of the process by which the gas spread to all parts of the laboratory?
(iii) State why first aid kit should be easily accessible in the laboratory

(b) State three disadvantage of using hard water in washing and bathing.

5. (a) Mellina has been instructed by her teacher to make sodium chloride solution by adding hydrochloric acid to sodium hydroxide solutions
   (i) Identify the process taking place in this reaction
   (ii) How could mellina decide when she had added enough hydrochloric acid to react with sodium hydroxide solution since both sodium hydroxide and hydrochloric acid are colorless solution
   (iii) Write all requirements and explanation on how solid sodium chloride solute can be obtained from the solution
   (b) Why is it difficult to prepare hydrogen gas using aluminum and magnesium

6. (a) A student dropped few pieces of marble chips in dilute hydrochloric acid contained in a test tube The gas involved was passed through lime water. As a form three student
   (i) State the changes observed in lime water
   (ii) Write all chemical reactions in both cases
   (b)Why does a wall not immediately acquire a white colour when a coating slaked lime is applied (use support chemical equation in your explanation)
   (c)Explain clearly why boiling method is not suitable for removing permanent hardness of water

7. (a) What volume of oxygen gas measured at S.T.P would be required to combust all the hydrogen produced by treating 6 grams of magnesium with excess dilute sulphuric acid
   (b) Give detailed information on the following statements
      (i) Fire caused by electricity is not given their own class of fire
      (ii) Water is not suitable for extinguishing fire caused by flammable liquids
      (iii) Hydrogen is not often found free on its own on earth surface instead it is found in combination with many other elements.

8. (a)A certain commercial sulphuric acid solution contains 45% acid by mass and is packed in 2.5 litres bottles. If the density of such acids is 1.48g/cm^3. Calculate
      (i) Mass of sulphuric acid present in the bottle
      (ii) Molarity of commercial acid
      (iii) Volume of the acid required to be diluted into 2.00 litres of 0.05M of the acid solution

   (b) You are provided with element A and B with atomic number 20 and 17 respectively
      (i) Write an electronic configuration of the element A and B
      (ii) Write the formula of the compound formed when element A combine with element B and state the type of bond formed
      (iii) State two properties of the bond formed between A and B

SECTION C (30 Marks)

Answer **all** questions in this section

9. We have coal at Iringa region. Authorities in the government have allowed the use of coal for domestic and industrial purpose. What warning you can rise concerning likely effects. Give five points
10. Sodium chloride is known with different names such as table salts and common salts. Its used at home for adding flavor to food and other many uses, apart from sodium chloride there are other many kinds of salts. Explain six of them and their uses
11. Due to development of science and technology has led to discoveries of various chemical substance which are very useful in our daily life. Misuse of the chemicals discoveries might cause dangers in our daily life. With examples in each name five chemicals substance, their warning sign and the precaution that should be taken when using it

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